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Capturing Clay-CO₂ Interactivity via Water Analysis for CO₂ Storage Applications

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Abstract

This study investigates the effect of brine/CO₂ interactions with clay-rich (montmorillonite-rich, and illite-rich) formations, via water analysis using Capillary Electrophoresis. When considering CO₂ storage in clay rich reservoirs, one needs to consider the reaction between clay, brine, and CO₂. This reaction is complex and controlled by many factors, including the brine salinity and its ionic composition. The reaction between clay/brine/CO₂ can lead to dissolution of clay minerals, which leads to changes in the reservoir's physical properties (e.g., porosity and permeability), impacting the reservoir's storage capacity. In this study, two clay samples were considered, montmorillonite and illite. The clay samples, of known volume, were added separately to deionized water and NaCl brines of different salinities (i.e. 0, 5, 30, 40, 50, and 60 kppm), with and without dissolved CO₂. The clay samples were given 24 hr to interact spontaneously in the various fluids. The clay/fluid and CO₂/clay/fluids interactions were captured and analyzed using capillary electrophoresis method.

The analysis results demonstrated that montmorillonite and illite are reactive with both DI water and NaCl brines with different salinities. We observed that the lower the brine salinity, the higher the interaction/dissolution of the clay minerals was. In addition, clay mineral dissolution in the presence of CO₂ was more pronounced for both types of clays because of the formation of carbonic acid which enhances the clay mineral dissolution. Montmorillonite demonstrated greater reactivity than illite because it has greater cation exchange capacity, greater surface area as well as higher swelling potential. The results and findings of this study offer important information for the prediction of long-term CO₂ storage in montmorillonite-rich and illite-rich reservoirs.

Introduction

Carbon dioxide capture and storage (CCS) is a process that separates the CO₂ from industrial and energy related sources and transports it to a storage location as a long-term isolation from the atmosphere (Siqueira et al 2017), such as in depleted oil and gas reservoirs (Holloway and van der Staaten, 1995).

One of the challenges with CO₂ storage is that CO₂ injection can cause several physical and chemical changes to the geological media, affecting both reservoir and caprock for large distances and long periods of

time (Gaus 2010). These changes happen quickly in carbonate media and might have big impacts during the injection and storage processes (Rötting et al 2015). As carbon dioxide is injected, it leads to a disequilibrium of the fluid chemistry and pore pressures inside the reservoir and the caprock. In the surroundings of the injection wells, CO₂ dissolves in the reservoir fluids and lowers the pH. This causes mineral dissolution and alteration of the porosity and permeability (Bacci et al 2011). Furthermore, CO₂ dissolves very fast in the formation water - usually from minutes to hours - and quickly lowering the pH of the aqueous phase and as a result, changes the distribution of a lot of dissolved species (Gaus 2010). In carbonate rock reservoirs, this will cause the aqueous phase to be under-saturated with regard to carbonate minerals, which will lead to the dissolution of calcite, dolomite, and magnesite. With sufficient dissolution, the pore structure of the rock that regulates fluid flow will change structurally leading to changes in mechanical properties, residual trapping, absolute and relative permeability, and porosity (Bemer et al 2010).

The well-known chemical reactions initiated by dissolving CO₂ in reservoir aqueous fluids are described by Ortoleva et al., 1998. Firstly, CO₂ dissolution in water forms carbonic acid which significantly acidifies the medium.



Then, carbonic acid dissociates, forming bicarbonate ions.



Many reservoirs contain clay minerals like kaolinite, illite, montmorillonite, muscovite and chlorite in abundance. Clay is highly sensitive to low pH fluids and fresh water (El-Din Mahmoud, 2018). Water is adsorbed in the interlayer spaces of some clay minerals and interferes with clay quantitative analysis (Bergaya and Lagaly, 2006). Furthermore, when saturated with water, montmorillonite contains two planes of water molecules in the interlayer space.

According to Aksu et al 2015, the formations most vulnerable to damage are those with a large number of clays in their pores, such as kaolinite, montmorillonite, chlorite and illite. The influence of these clay minerals on the quality of the reservoirs is complex because of the reservoir's heterogeneity. The quality of the reservoirs is directly related to the structure of its pore space, and it is affected by many factors such as mineralogy of the rock and the impact of injected water on swelling of clay minerals (Sharifipour et al., 2018). Furthermore, the distribution and concentration of clay minerals in the pores of rock greatly impacts the permeability of the reservoirs (Ahmad et al. 2018).

In the case of clay-containing reservoirs, the interactions between clay/brine/CO₂ are complex and can significantly influence the reservoir's physical properties. The reaction between clay minerals and CO₂-saturated brine can lead to dissolution of these minerals. This process can alter the reservoir's porosity and permeability, which are critical for CO₂ storage capacity. In addition, the salinity and composition of the formation brine play a significant role in this reaction.

In this study, the effect of CO₂ on montmorillonite-rich, and illite-rich formations with different fluids including DI water and NaCl brine with different salinities, was studied, through detailed experimental procedures. The outcome of the study helps to further understand the implications of storing CO₂ into clay-rich formations and to predict the long-term reservoir stability and ability to store CO₂.

Experimental Materials

Clay samples

Two types of clay minerals were selected for the study: montmorillonite, and illite. These clay samples were obtained from WARD's Natural Science. These different clay types have distinct chemical formulas: montmorillonites has the chemical structure of (Na,Ca)_{0.3}(Al,Mg)₂Si₄O₁₀(OH)₂.nH₂O and it is well known of its higher swelling tendency and higher cation exchange capacity (Foster, 1953). Illite's structure is mainly

composed of alumina, silica, and water, with other associated elements like potassium, magnesium, and iron, and given by the formula: $(K,H_3O)(Al,Mg,Fe)_2(Si,Al)_4O_{10}[OH]_2 \cdot H_2O$ (Wilson, 2004; Deer et al., 1991). Illite has a high water absorption capacity (Supandi et al., 2019).

Fluids

Deionized (DI) water and NaCl brine solutions with different salinities of 0, 5, 30, 40, 50, and 60 kppm were used in this study. The carbonated brine was prepared by dissolving CO₂ gas into the different brine solutions.

Experimental procedure

2 wt% from each clay minerals were added separately to DI water and the different brines. The clay minerals were given a total time of 24 hrs to react with the brine, after which the solutions were filtered. The filtered solution was then collected and analyzed using CE, to capture the clay/brine interactions, if any, and these measurements were considered as baselines.

Water Analysis via Capillary Electrophoresis (CE)

To capture the clay/fluid interaction, water effluents were analyzed using a CAPEL-205 capillary electrophoresis unit equipped with a UV/VIS detector supplied by LUMEX. First, capillary conditioning was performed, which is a fundamental step before performing analysis in capillary electrophoresis. Proper conditioning ensures the capillary is clean, free from contaminants, and ready for accurate and reproducible results. This step consisted of an initial cleaning with (NaOH, deionized water), followed by the injection of the cation buffer solution for 5 minutes, and the last step was cleaning the capillary with deionized water. Once the capillary was cleaned, the analysis of the samples was done using a cation program set in the ELFORUN software. The maximum acceptable concentration of the NaCl sample by CE is 2000 ppm, therefore, NaCl brine samples with higher salinities were diluted before analysis, using deionized water.

To perform the measurement on the collected water effluents, the capillary was rinsed with an initiator and a cation separation buffer, after which the sample was injected into the capillary for measurement. A voltage of 30kV was applied across the capillary to perform the separation and the measurement was performed at a wavelength of 200 nm and at a constant temperature of 25 °C. Based on the chemistry of the samples, the analytes migrated at different migration times. In this study, only cations are analyzed, as the focus is to investigate the cation dissolution/interaction of clay-exposed to CO₂. Because the analysis was done with cation buffer, the CO₂ signal would not appear in electropherogram.

Results and Discussion

In clay rich formations, capturing clay/fluid interactivity is crucial for CO₂ storage projects. Clays tend to react differently with water and brine, depending on the salinity and composition, such that when clay is exposed to fresher water, it has the tendency to release ions, migrate and/or to swell. CO₂ injection and storage can induce mineral reactions, such as dissolution and precipitation, as CO₂ tends to solubilize in the brine, forming a low pH carbonic acid. Such things can influence CO₂ storage and geological sequestration in clay-rich formations. The following sections show our experimental results related to the interactions of CO₂/fluid/clay for different types of clay and fluid systems.

Analysis of the Pure Fluids

First, a baseline for subsequent analyses was executed on DI water and pure NaCl brines with a cation buffer using CE. The electropherogram of DI water did not show any peak, confirming the purity of the DI water, Figure 1. For 2,000 ppm NaCl brine, the electropherogram showed a clear peak corresponding to the presence of sodium ion Na⁺, at a migration time of 2.9 min, Figure 1.

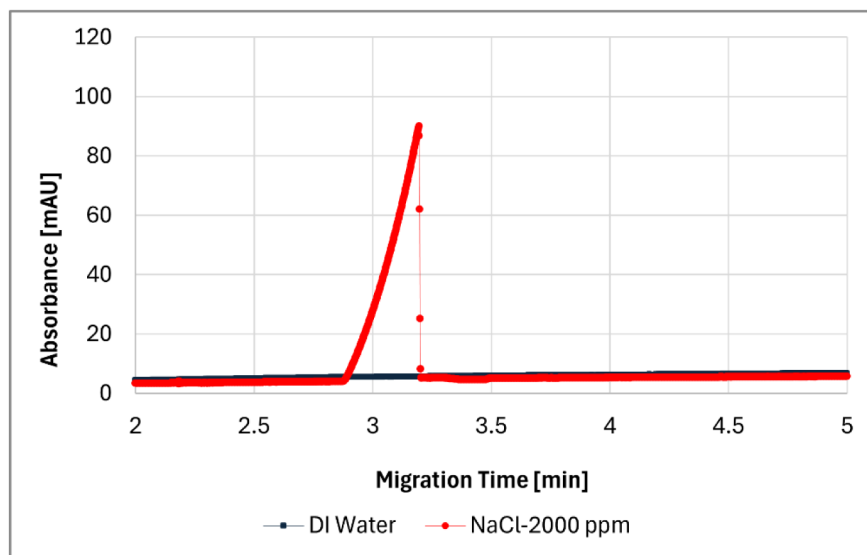


Figure 1—Electropherogram for DI water and NaCl brine (2000 ppm).

Then, the prepared carbonated DI water and the carbonated NaCl brine were also analyzed. The carbonated DI water was analyzed with the cation buffer, following similar procedures described earlier, and the result showed that the carbonated DI water had identical electropherogram as the one for DI water, Figure 2. Similarly, the carbonated NaCl brine gave identical electropherogram to the NaCl brine one. Therefore, any additional peaks, that appear after CO₂/fluid/clay experiments can be confidently assigned to the presence of dissolved clay cations, resulting from clay-interactions with the fluid medium.

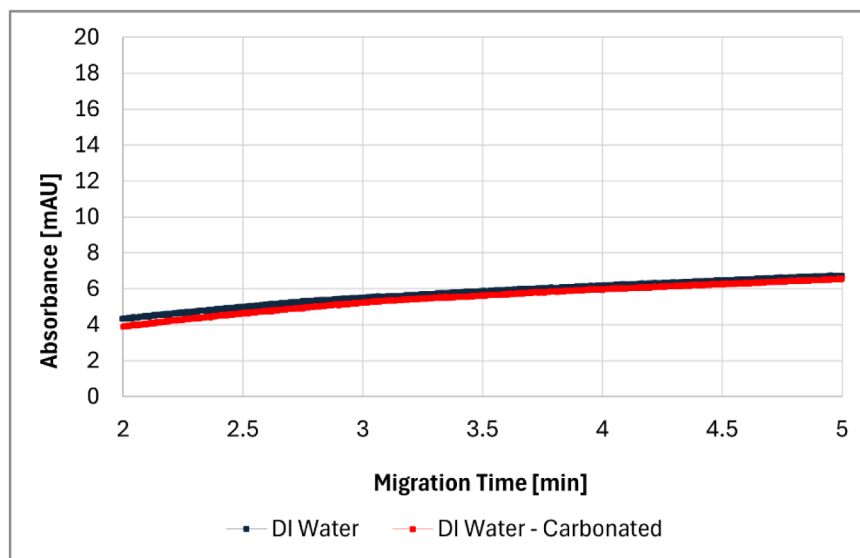


Figure 2—Electropherogram for carbonated DI water in comparison to DI water.

Analysis of Clay/CO₂/Fluids Interactions

Montmorillonite. After determining the baseline, the interactions of Clay/CO₂/Fluids were then investigated. Figure 3 compares the electropherograms of pure DI water vs. the effluents of a DI water solution exposed to 2 wt% montmorillonite. New peaks appear in the electropherogram, related to cations resulting from the dissolution of chemical elements from the clay. Montmorillonite with its two outer tetrahedral silicate layers and a middle octahedral alumina layer, can exhibit significant potential reactivity

and swelling, when interacting with water (Brindley, 1979). When montmorillonite absorbs water, its interlayers cations (sodium, potassium, calcium, and magnesium) can be exchanged with water molecules. This dissolution process of montmorillonite leads to an increase of these cations' concentration in the water (Sendula et al., 2017), which is observed in Figure 3, where four new cation peaks were reported.

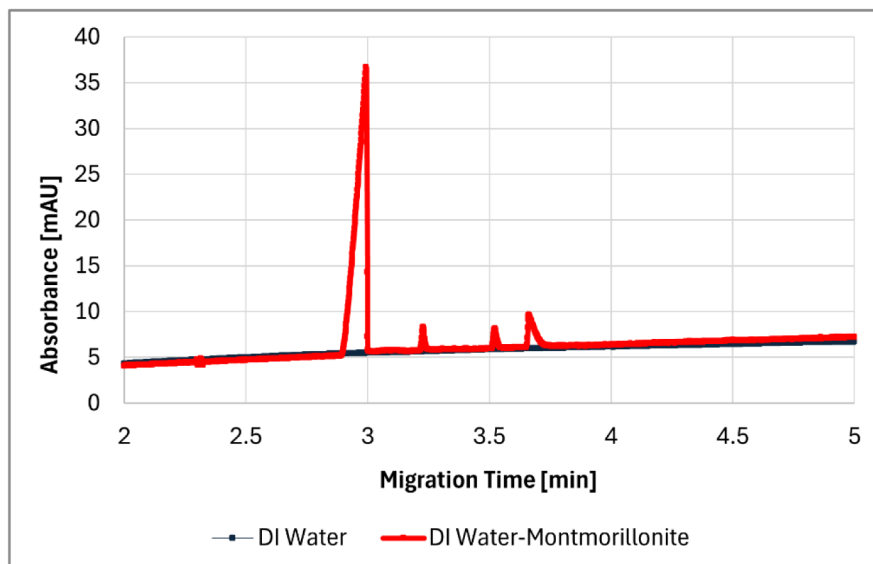


Figure 3—Electropherogram of DI water/montmorillonite in comparison to DI water baseline.

Whilst investigating the effect of CO_2 on montmorillonite, we observed larger peaks on the electropherograms, indicating a more pronounced interaction between the carbonated water and the clay (Figure 4). This can be explained by the increase of the acidity of the solution. When CO_2 is dissolved in the water, it forms carbonic acid which could enhance the mineral/clay dissolution (Fani et al., 2024).

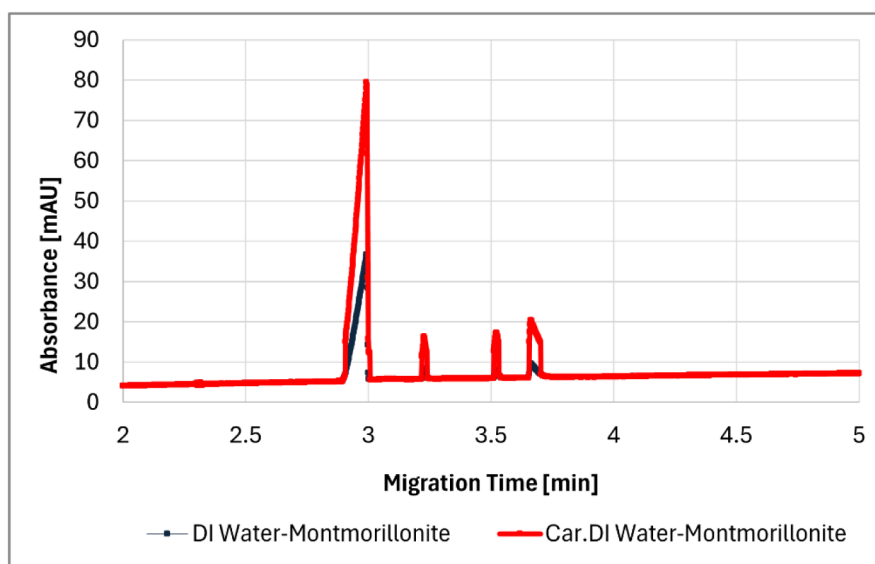


Figure 4—Electropherogram of carbonated DI water/montmorillonite in comparison to DI water/montmorillonite.

The effect of varying brine salinities was also investigated, as part of this study. Figure 5 gives an example of an electropherogram for a 5kppm brine. At such low salt concentration, the ions dissociation is expected especially in the presence of monovalent cations like the Na^+ (Dashtian and Bakhshian, 2021). The area

under the new peaks were integrated with the exception of the Na^+ peak, as it overlaps with the Na^+ peak from the baseline.

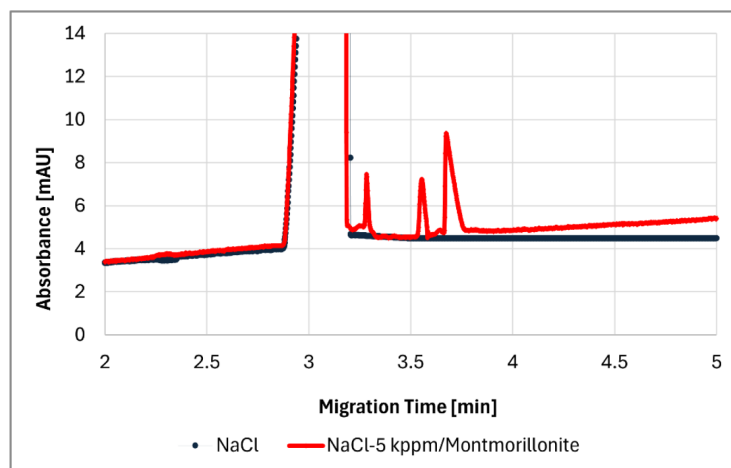


Figure 5—Electropherogram of NaCl-5kppm/montmorillonite in comparison to NaCl baseline.

The 5 kppm NaCl brine appears to be more reactive than DI water only, as demonstrated by the difference in amplitude of the integrated peaks (140 mAU and 72 mAU, respectively for 5 kppm NaCl brine and DI water). This is expected, especially at this range of brine salinity. When NaCl dissolves in water, it introduces Na^+ ions, which can interact with the clay and change its surface properties. In DI water case, however, the clay is mainly interacting through hydrogen bonding and dipole-dipole interactions, which are of lesser impact compared to Na^+ (Johnston, 2018).

As the salinity of the NaCl brines increases, the dissolution/interaction of montmorillonite decreased, as can be seen in Figure 6 (solid marker). At higher brine salinity, the electrical double layer, surrounding the clay particles, will be squeezed, resulting in a repulsive force that hinders clay's ability to adsorb water and interact (Behbahani et al., 2021; DeCarlo and Shokri, 2014), which could explain the lower reaction intensity beyond 30 kppm.

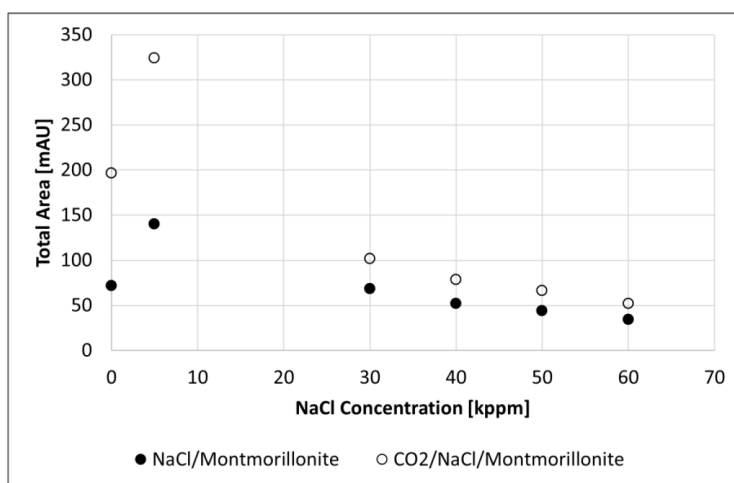


Figure 6—Total area for montmorillonite in DI water, carbonated DI water, NaCl brines and carbonated NaCl brines at different salinities.

For the case of carbonated brines, however, the interaction of the clay was found to be more pronounced, where higher peak amplitudes and higher integrated areas were reported (example is illustrated in Figure 7).

This, as explained earlier, is due to the presence of dissolved CO₂ which changes solutions' chemistry and reactivity, resulting in more ions dissociations (Jo et al., 2012; Hidayat et al., 2022). The brine interaction with CO₂, produces carbonic acid, which lowers the pH and enhances the dissolution of montmorillonite, releasing cations into the brine solution.

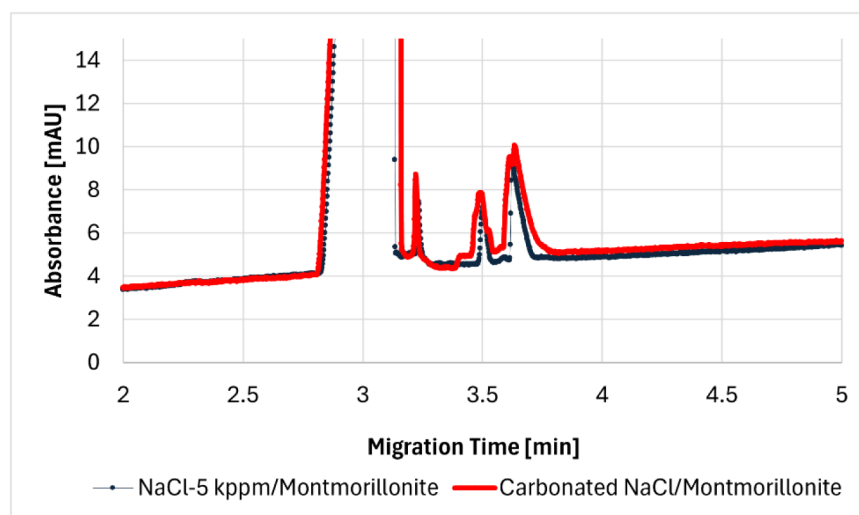


Figure 7—Electropherogram of NaCl-5kppm/montmorillonite in comparison to NaCl-5kppm/CO₂/montmorillonite.

Similar to the pure NaCl brine case, the clay reaction intensity with carbonated brine was also found to decrease after 30 kppm, as can be seen in Figure 6 (empty marker). For similar reason discussed earlier, the Na⁺ peak was not considered in the peaks' area integration. The calculated areas are summarized in Table 1.

Table 1—Total calculated areas for montmorillonite electropherograms in DI water and NaCl brines.

Effluent	Integrated Areas [mAU]	Effluent	Integrated Areas [mAU]	Ratio of Integrated Areas
DI water	72.00	Car.DI water	196.3	2.73
5 kppm NaCl	140.00	Car. 5 kppm NaCl	324.12	2.32
30 kppm NaCl	68.60	Car. 30 kppm NaCl	101.70	1.48
40 kppm NaCl	52.10	Car. 40 kppm NaCl	78.43	1.51
50 kppm NaCl	44.10	Car.50 kppm NaCl	66.35	1.50
60 kppm NaCl	34.40	Car.60 kppm NaCl	52.00	1.51

Figure 8 compares the integrated areas of montmorillonite/CO₂/fluid with montmorillonite/fluid. The ratios of the integrated areas for montmorillonite/CO₂/fluid were found to be 2.73 and 2.32 times higher than that for montmorillonite/fluid, for DI water and 5 kppm NaCl, respectively. For higher salinity solutions, however, 30 kppm and higher, the ratios of montmorillonite/CO₂/fluid to montmorillonite/fluid were found to be 1.50 across all tested salinities, above 30 kppm. These results confirm the tendency of CO₂ to result in significantly higher clay/fluid interactions, even at higher salinities.

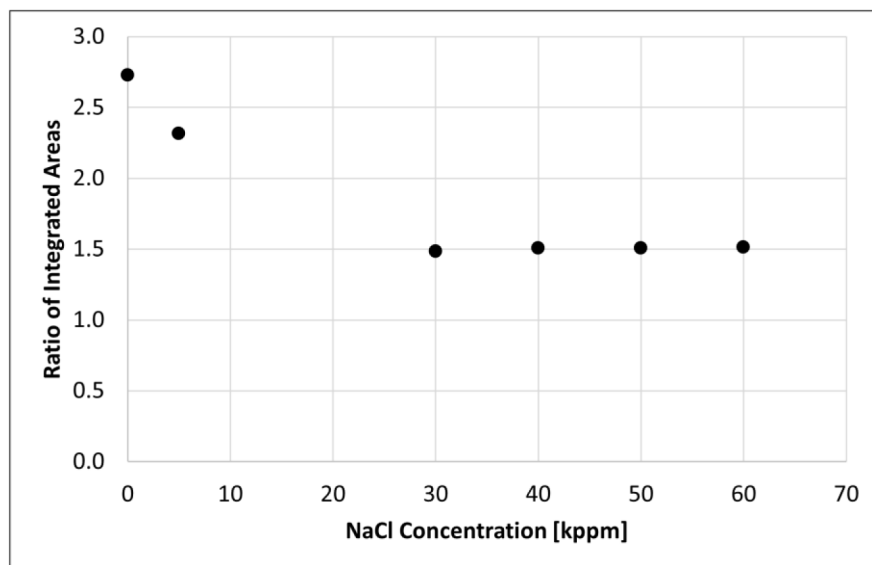


Figure 8—Ratio of integrated areas of montmorillonite/CO₂/fluid to montmorillonite/fluid.

Illite. Another type of clay, i.e. illite, was considered and Clay/CO₂/Fluids interactions were investigated, following similar procedures described earlier. The electropherogram of the DI water/illite interaction is shown in Figure 9, where clear interaction/dissolution can be noted. Three new peaks appeared in the electropherogram. Illite with its two inward-pointing silica tetragonal sheets and its central octahedral sheet can react with water, by which it releases dissolved cations; potassium (K⁺), aluminum (Al³⁺) and silicon (Si⁴⁺) (Muraleedharan et al., 2020). Using the ELFURON software, the area under the peaks was calculated and found to be 49.9 mAU, and this will be considered as a baseline for the further discussed clay/fluids interactions.

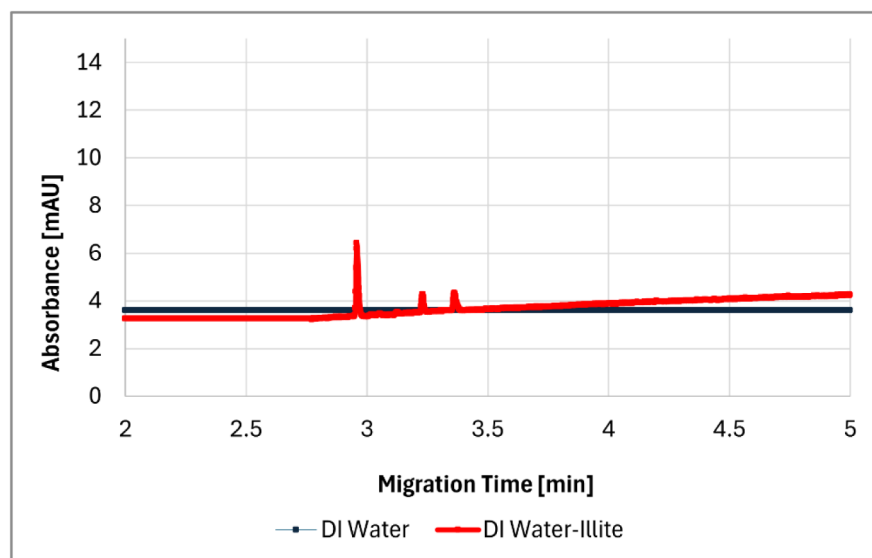


Figure 9—Electropherogram of DI water/illite in comparison to DI water baseline.

Then, the effect of CO₂ was investigated, by studying the reaction of CO₂/DI water/illite. The electropherogram result is shown in Figure 10, where higher peaks intensity, implying higher interactivity, was observed compared to the case of DI water/illite case. This increase in reactivity is due to the presence

of dissolved carbon dioxide in carbonated water, which can form a low pH carbonic acid. This acid, with its low pH, can enhance further the dissolution of illite minerals (Gaus 2010, Bacci et al 2011).

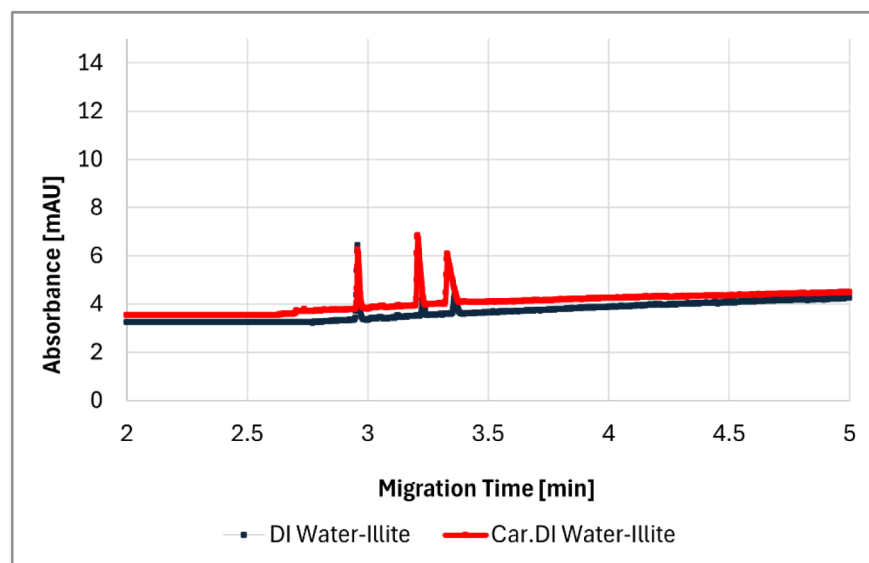


Figure 10—Electropherogram of carbonated DI water/illite in comparison to DI water/illite.

Then, illite/brine interaction was studied at different brine salinities of 5, 30, 40, 50 and 60 kppm. Example of the resulting electropherogram is shown in Figure 11, for 5 kppm case. Illite shows to interact with NaCl brines across the various salinities. The reaction intensity, however, found to decrease at higher brine salinity, as shown in Figure 12 (solid marker). The low reaction at higher salinity, can be due to the compression of the electrical double layer around the clay particles, as previously mentioned. Table 2 summarizes the total calculated areas under the generated peaks, for illite interaction in DI water and NaCl brines.

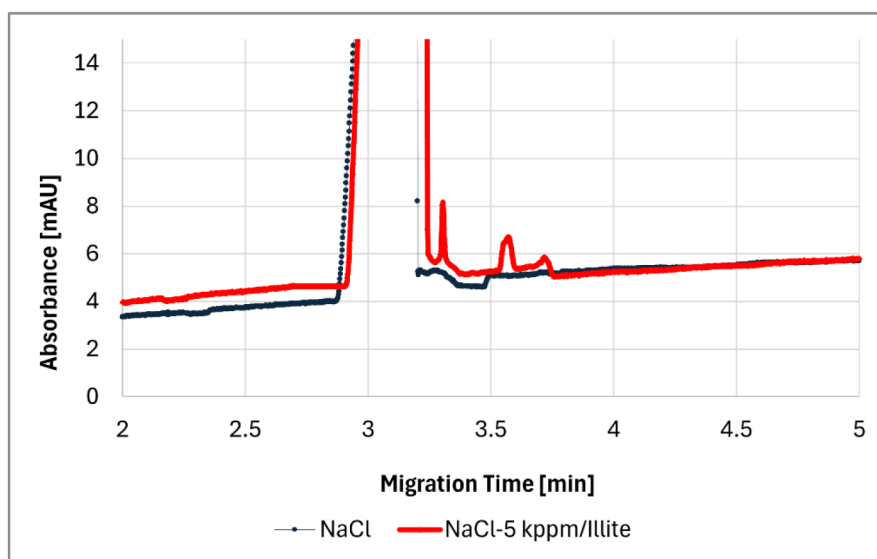


Figure 11—Electropherogram of NaCl-5kppm/illite in comparison to NaCl baseline.

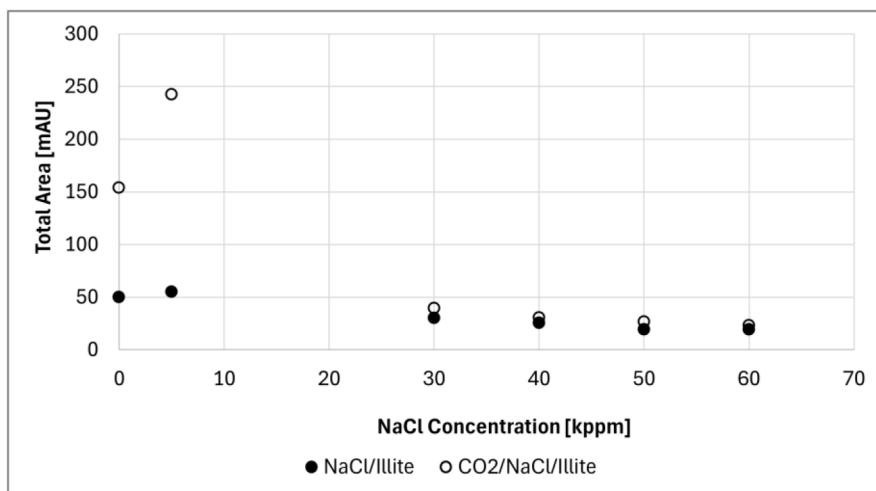


Figure 12—Total area for illite in DI water, carbonated DI water, NaCl brines and carbonated NaCl brines at different salinities.

Table 2—Total calculated areas for illite electropherograms in DI water and NaCl brines.

Effluent	Integrated Areas [mAU]	Effluent	Integrated Areas [mAU]	Ratio of Integrated Areas
DI water	49.93	Car.DI water	153.74	3.08
5 kppm NaCl	55.04	Car. 5 kppm NaCl	242.37	4.40
30 kppm NaCl	29.94	Car. 30 kppm NaCl	39.61	1.32
40 kppm NaCl	25.28	Car. 40 kppm NaCl	30.38	1.20
50 kppm NaCl	19.35	Car.50 kppm NaCl	26.70	1.38
60 kppm NaCl	19.36	Car.60 kppm NaCl	23.12	1.19

Overall, we observe a lesser effect of the salt concentration in the brine on the rate of interaction between illite and the solution, compared with the observations we made for montmorillonite. As an example, the peak intensities for the reaction of illite with DI water and NaCl brine are very similar (49.93 mAU and 55.04 for DI water and 5 kppm NaCl brine, respectively), indicating comparable rates of reaction.

Then, the CO₂/brine/illite interactivity was investigated, where higher reaction intensity was noted compared to those with pure NaCl brines (an example is shown in Figure 13 for 5 kppm NaCl. The presence of CO₂ enhances the clay interactivity with water, as it lowers the pH and therefore enhances the dissolution of illite.

Comparing the interactivity of CO₂/brine/illite at different brine salinities, similar observation to the NaCl brine case can be noted (Figure 12). The lower the brine salinity, the higher the dissolution of illite minerals. This is, as explained earlier, due to the lower ionic strength, which provide fewer ions to compete with mineral's surface site, allowing more effective interaction between illite and low salinity brine.

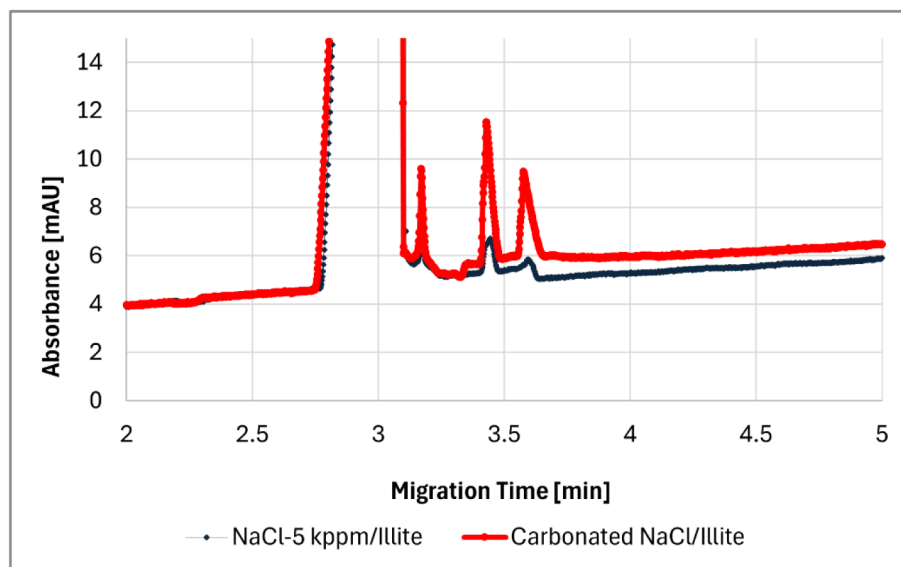


Figure 13—Electropherogram of NaCl-5 kppm/illite in comparison to NaCl-5kppm/CO₂/illite.

Comparing the interactions of illite/CO₂/fluid and illite/fluid, it can be again noted that the presence of CO₂ significantly affects the illite/fluid reactivity. At low salinity conditions, the ratio of the integrated areas for the illite/CO₂/fluid was found to be 3.08 and 4.40 higher compared to illite/fluid, for DI water and 5 kppm NaCl brine, respectively. For higher salinities, >30 kppm, the presence of CO₂ still showed higher interactions tendency. The interactions tendency at higher salinity, however, was less pronounced compared to DI water and 5 kppm NaCl brine. The ratios were fluctuating between 1.19 - 1.38 across the salinities between 30-60 kppm, as shown in Figure 14.

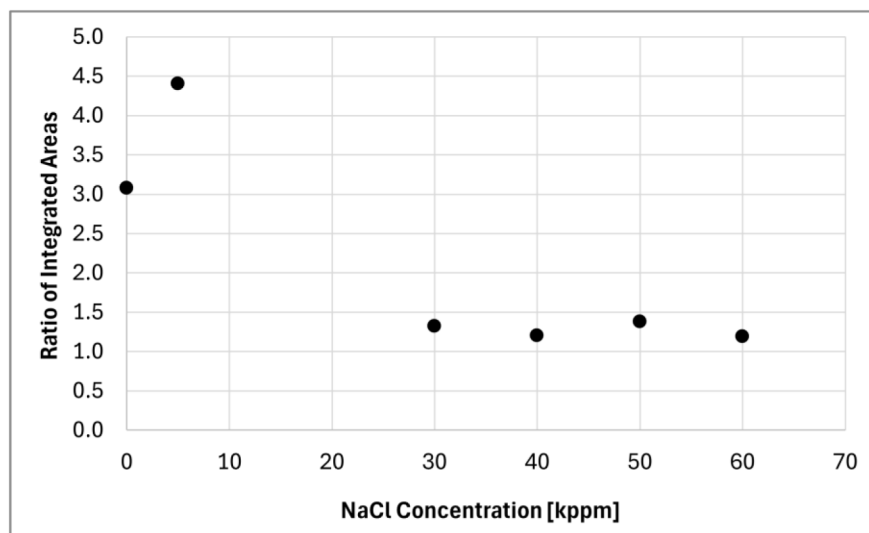


Figure 14—Ratio of integrated areas of illite/CO₂/fluid to illite/fluid.

Comparison of Montmorillonite and Illite Interaction with CO₂. Comparing the interactivity of both studied clay types, montmorillonite and illite, with the studied fluids, it can be noted that montmorillonite was more reactive. Montmorillonite with its higher cation exchange capacity, greater surface area and higher swelling potential, showed significantly more reactivity to the various NaCl salinities, compared to illite (Figure 15 (a)). Both clays were reactive with all the NaCl brines studied, even at the highest studied salinity of 60 kppm. In all cases, the presence of CO₂ showed to enhance further the clay/fluids interactivity. Again,

the montmorillonite, however, was found to be more reactive at the presence of CO_2 compared to illite (Figure 15 (b)). Such more reactivity of montmorillonite is due to its lower layer charge and the presence of readily ionizable water molecules within its interlayer space. These allow much more swelling tendency and higher cation exchange capacity, compared to illite (Overmann et al., 2024).

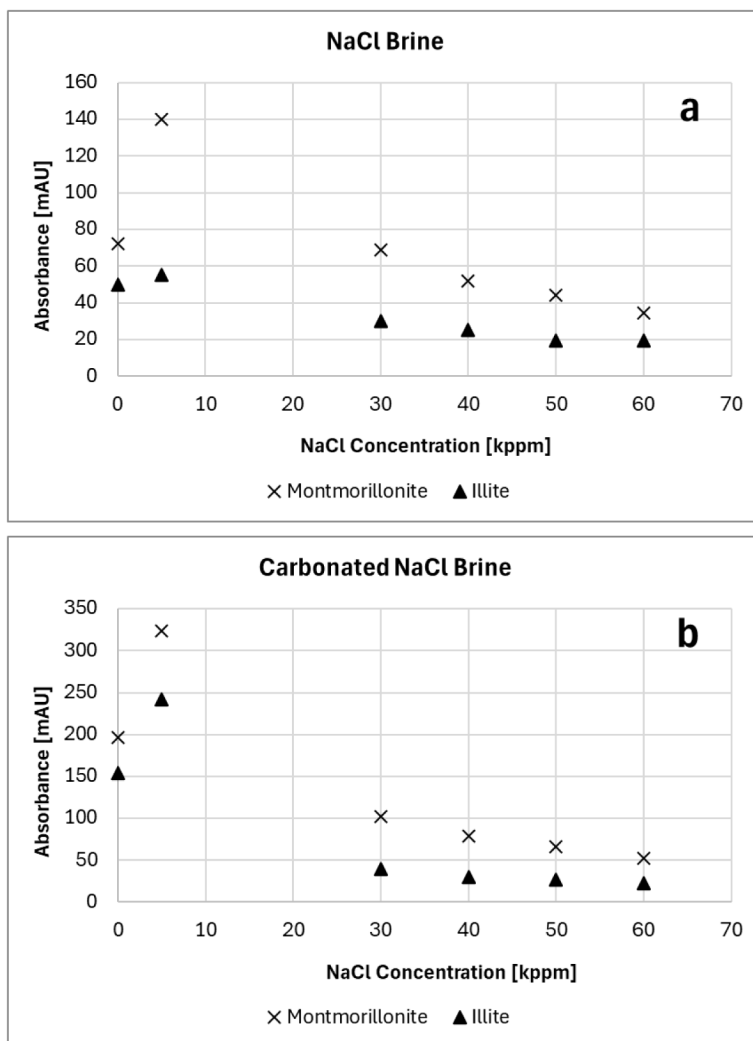


Figure 15—Comparison between montmorillonite and illite reactivity at the various studied fluids (a) in NaCl brine; (b) in carbonated NaCl brine

Conclusion

The reactivity of montmorillonite and illite clays with CO_2 was investigated in this study. CO_2 was dissolved in deionized water and NaCl brines of various salinities, separately. Then known clay concentration was allowed to spontaneously react with the various fluids, for 24 hr, with and without the presence of CO_2 . The reactants effluents were collected and analyzed by capillary electrophoresis to capture the fluids/clay and CO_2 /fluids/clay reactivity. The results showed that both clays are reactive with deionized water and NaCl brines over the range of salinities studied. The reaction intensity with the brines, however, was found to decrease at higher salinities, beyond 30 kppm.

The presence of CO_2 in the fluids studied was found to systematically enhance the reactivity of both clays. Dissolving CO_2 in aqueous solutions, forms a low pH carbonic acid, which promotes the dissolution and alteration of the clay mineral.

In all cases, the montmorillonite clay showed to be more reactive than illite, due to its higher cation exchange capacity, greater surface area and higher swelling potential. The outcomes of the study provide a good reference dataset for CO₂ storage in clay-rich reservoirs.

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